Execution-aware PPO for stateful contact control

R3 source-grounded implementation; performance advantage is not implied.

next observed state; no future-success input

Observed state + Actor–Critic

87D: pose, velocity, goals and geometry;
response history + controller memory.
5D: XY residual, contact, yaw, approach.
MLP 64 × 64; standard PPO + GAE.

Shared stateful executor

Nominal feedback + DLS servo.
Hold / refresh legal parameters.
Controller-owned geometric recontact.
No learned high-level stage switch.

PhysX + independent checks

One 16-env PhysX app, 120 Hz.
Free workpiece; no pose overwrite.
Observed response feeds next state.
Sampled geometry / tool checks,
not hardware safety or field transfer.

Execution-active likelihood

$$\ell_t = \sum_j m_{tj} \log \mathcal{N}(a_{tj}; \mu_{\theta j}, \sigma^2_{\theta j})$$

Pre-action mask is in the observation.
Inactive coordinates integrated out.
Masked log-probability enters ratio.
Entropy coefficient = 0.

Implemented mechanism only; success gain and seed robustness require independent evidence.
event_qualified_seed31 | source e31be3c1bc94 | 32,768 interactions
Not a new PPO theorem. Not field / hardware transfer. Contract and execution traces retained.